

Assignment 2

1. Solve the following first order ordinary differential equation for $y(1.2)$ and $y(1.4)$ using 4th order Runge-Kutta method:

$$\frac{dy}{dx} = \frac{2xy + e^x}{x^2 + xe^x}; \quad y(1) = 0.$$

2. Solve the following first order ordinary differential equation for $y(0.2)$ using 4th order Runge-Kutta method:

$$\frac{dy}{dx} = \frac{y - x}{y + x}; \quad y(0) = 1.$$

3. Solve the following first order ordinary differential equation for $y(0.1)$ and $y(0.2)$ using 4th order Runge-Kutta method:

$$y' + 2y = x^3 e^{-2x}; \quad y(0) = 1.$$

4. Solve the following first order ordinary differential equation for $y(4.2)$ and $y(4.4)$ using 4th order Runge-Kutta method:

$$\frac{dy}{dx} = \frac{1}{x^2 + y}; \quad y(4) = 2.$$

5. The following data gives the velocity of a particle for twenty seconds at an interval of five seconds. Find the initial acceleration using the entire data:

Time t (sec):	0	5	10	15	20
Velocity v (m/sec):	0	3	14	69	228

6. Find the value of $\cos(1.74)$ from the following table:

x :	1.7	1.74	1.78	1.82	1.86
$\sin x$:	0.9916	0.9857	0.9781	0.9691	0.9584

7. Evaluate the following integral using Simpson's 1/3 rule.

$$\int_0^1 \frac{x^2}{1+x^3} dx$$

Compare the error with the exact value.

8. Use the Simpson's 1/3 rule to estimate the following integral by taking seven ordinates.

$$\int_0^{0.6} e^{-x^2} dx$$

9. Use the Trapezoidal rule to estimate the following integral by taking the number of 10 sub-intervals.

$$\int_0^2 ex^2 dx$$

10. Calculate the approximate value of I by using Trapezoidal rule, Simpson's 1/3 rule and Simpson's 3/8 rule. Find the Exact Value of I , then Compare and Comment on the results.

$$I = \int_{0.2}^{2.4} \frac{(1+x)e^x}{\cos^2(xe^x)} dx$$

11. Evaluate the following integral using any two numerical integration methods with nine equidistant ordinates. Then, compare the results and give your opinion on which method is better and why.

$$\int_0^8 \frac{1}{x+1} dx$$

12. Calculate the approximate value of I by using Trapezoidal rule, Simpson's 1/3 rule and Simpson's 3/8 rule. Find the Exact Value of I , then Compare and Comment on the results.

$$I = \int_0^{\pi} \frac{\sin x}{2 + \cos x} dx$$